16 Here are the first six terms of a quadratic sequence.

−1 5 15 29 47 69

Find an expression, in terms of n, for the nth term of this sequence.

First differences: 6, 10, 14, 18, 22.

1 mark
Constant second difference 4.

Second differences: 4, 4, 4, 4.

$2a = 4$, so $a = 2$; trial term $2n^2$.

1 mark
Leading term $2n^2$.

At $n = 1$, $2n^2 = 2$ but term = −1: subtract 3.

nth term = $2n^2 - 3$.

1 mark
$2n^2 - 3$.

$2n^2 - 3$

(Total for Question 16 is 3 marks)